

Question number	Answer	Additional guidance	Mark
3(a)	The correct answer is C <i>A is incorrect because active transport can move solutes against the concentration gradient</i> <i>B is incorrect because active transport can move solutes against the concentration gradient</i> <i>D is incorrect because water moves in osmosis not solutes</i>		(1)

Question number	Answer	Additional guidance	Mark
3(b)	The correct answer is A <i>B is incorrect because only active transport involves proteins</i> <i>C is incorrect because only active transport involves proteins</i> <i>D is incorrect because only active transport involves proteins</i>		(1)

Question number	Answer	Additional guidance	Mark
3(c)	The correct answer is B <i>A is incorrect because active transport does not move large particles or bacteria</i> <i>C is incorrect because exocytosis moves matter out of the cell</i> <i>D is incorrect because facilitated diffusion does not move large particles or bacteria</i>		(1)

Question number	Answer	Additional guidance	Mark
3(d)	The correct answer is B <i>A is incorrect because lipid-soluble molecules can move directly through the membrane by diffusion</i> <i>C is incorrect because lipid-soluble molecules can move directly through the membrane by diffusion</i> <i>D is incorrect because osmosis is the movement of water</i>		(1)

Question number	Answer	Additional guidance	Mark
3(e)	The correct answer is B <i>A is incorrect because polar R groups cannot interact with the phospholipid tails</i> <i>C is incorrect because polar R groups cannot interact with the phospholipid tails and non-polar R groups cannot interact with the polar molecules passing through the channel</i> <i>D is incorrect because polar R groups are needed to allow polar molecules through</i>		(1)

Question number	Answer	Additional guidance	Mark
3(f)	An explanation that includes four of the following points: • (change in mass / movement of water) is due to <u>osmosis</u> (1) • potato decreased mass in {purple / orange} solutions because the solutions were more concentrated than the {potato / cytoplasm} (1) • decrease in mass was greatest in orange solution as this was more concentrated than the purple solution (1) • potato increased in mass in the red solution as the concentration of the solution was less than that of the {potato / cytoplasm} (1) • no change in mass in blue solution as the concentration of the solution and the {potato / cytoplasm} were the same (1)	NB ACCEPT converse explanations Penalise once water concentration NB more concentrated solution = lower water potential lower solute potential lower osmotic potential less dilute hypertonic NB less concentrated solution = higher water potential higher solute potential higher osmotic potential more dilute hypotonic ACCEPT isotonic no <u>net</u> movement of water same water potential same solute potential same osmotic potential	(4)